Aptamer-gelatin composite for a trigger release system mediated by oligonucleotide hybridization.

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Nucleic acid aptamers not only specifically bind to their target proteins with high affinity but also form intermolecular hybridization with their complementary oligonucleotides (CO). The hybridization can interrupt aptamer/protein interaction due to the changes of aptamer secondary structure which rely on hybridization length and base-pairing positions. Herein we aim to use this unique property of the aptamers, when combined with gelatin to develop a novel composite with desirable protein release profiles. Platelet-derived growth factor-BB (PDGF-BB) and its aptamer were used as target molecules. Prior to performing the release study, the effects of CO on aptamer-protein interaction were observed by surface plasmon resonance (SPR). The SPR sensorgram indicated that the aptamer dissociated from the bounded proteins when it hybridized with the CO. The aptamer was then immobilized onto streptavidin coated polystyrene particles via biotin/streptavidin interaction. Then, PDGF-BB and aptamer functionalized particles were mixed with gelatin solution and cast as small pieces of composite. The success of the composite preparation was confirmed by flow cytometry and microscopy. PDGF-BB release at several time points was quantified by ELISA. The results showed that the aptamer-gelatin composite could slow the release rate of the proteins from the composite due to strong binding of proteins and aptamers. Once the CO was added to the system, the release rate was significantly enhanced because the aptamer hybridized with the CO and lost its active secondary structure. Therefore, the proteins were triggered to release out from the composite. This work suggests a promising strategy for controlling the release of bioactive molecules in medical treatments.

Emerging techniques employed in aptamer-based diagnostic tests.

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Since aptamers were reported in 1990, research into the applications of aptamers, particularly diagnostic applications, has been growing. Aptamers can act as recognition elements instead of antibodies. In this regard, aptamers have unique characteristics because they are composed of nucleic acids. Intra- and intermolecular interactions of nucleic acids can be easily tailored following straightforward hybridization rules. Nucleic acids can be enzymatically replicated and their sequences can be determined using high-throughput methods. Using these properties, ligand-induced structural change-based aptamer sensors for homogeneous assays, polymerase- and/or nuclease-combined aptamer sensors for ultrasensitive assays, and microarray/next-generation sequencing-based aptamer sensors for multiplexed assays have been developed. This article reviews these unique aptamer sensors, demonstrating their great potential for diagnostic applications.

Rabbit antibody detection with RNA aptamers.

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Antibody-based detection systems are widely used, but in the cases of immunoprecipitations and enzyme-linked immunoassays, they can be laborious. These techniques require the preparation of at least two kinds of non-cross-reactive immunoglobulin Gs (IgGs), usually made from different species against the single target molecule. Aptamers composed of nucleic acids possess strict recognition ability for the target molecule's three-dimensional structure and, thus, are considered to act like IgG. In this study, experimental trials were designed to combine the advantages of IgG and aptamers. For this purpose, aptamers against rabbit IgG were identified by in vitro selection. One of the obtained aptamers had a dissociation constant lower than 15 pM to the rabbit IgG. It bound specifically to the constant region of the rabbit IgG, and no binding was observed with mouse or goat IgG. Moreover, this aptamer recognized only the native form of rabbit IgG and could not bind the sodium dodecyl sulfate (SDS)-denatured form. These features show the advantage of using the aptamer as a secondary probing agent rather than the usual secondary antibodies.

Stable excess mortality in a multiple sclerosis cohort diagnosed 1970-2010.

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Multiple sclerosis (MS) is associated with excess mortality. The use of disease-modifying treatments (DMTs) has recently been associated with survival benefits. A regional MS database was linked with national registries. People with MS (pwMS) diagnosed in 1971-2010 were included and followed up until the end of the year 2019. Five matched controls were acquired for every person with MS. DMTs included in the analyses were interferon and glatiramer acetate. Median follow-up time of the 1795 pwMS was 20.0 years (range 0.1-48.7 years). Survival did not differ between decades of diagnosis ($p = 0.20$). Amongst pwMS, male sex (adjusted hazard ratio [aHR] 1.70; 95% confidence interval [CI] 1.41-2.06), higher age at diagnosis (aHR 1.83; 95% CI 1.65-2.03 per 10-year increment) and primary progressive disease course (aHR 1.29; 95% CI 1.04-1.60) were independently associated with poorer survival. DMT use was associated with better survival ($p < 0.0001$) and better survival during follow-up (aHR 0.56; 95% CI 0.38-0.81). Compared to matched controls, median life expectancy was 8-9 years shorter in pwMS with survival diverging from controls during the first decade after diagnosis, more clearly in men than women. Despite DMT use being associated with better survival, relative life expectancy of pwMS did not change over five decades in Western Finland. Male sex was an independent risk factor for death amongst pwMS, but excess mortality was higher in women. More work and methods are needed to improve survival in pwMS.

Multiple sclerosis mortality in New Zealand: a nationwide prospective study.

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Mortality data from Europe and North America show a shorter life expectancy for people with multiple sclerosis (MS). It is not known if a similar mortality risk exists in the southern hemisphere. We analysed the mortality outcomes of a comprehensive New Zealand (NZ) MS cohort, 15 years postrecruitment. All participants of the nationwide 2006 NZ MS prevalence study were included and mortality outcomes were compared with life table data from the NZ population using classic survival analyses, standardised mortality ratios (SMRs) and excess death rates (EDRs). Of 2909 MS participants, 844 (29%) were deceased at the end of the 15-year study period. Median survival age for the MS cohort was 79.4 years (78.5, 80.3), compared with 86.6 years (85.5, 87.7) for the age-matched and sex-matched NZ population. The overall SMR was 1.9 (1.8, 2.1)). Symptom onset between 21 and 30 years corresponded to an SMR of 2.8 and a median survival age 9.8 years lower than the NZ population. Progressive-onset disease was associated with a survival gap of 9 years compared with 5.7 years for relapsing onset. The EDR for those diagnosed in 1997-2006 was 3.2 (2.6, 3.9) compared with 7.8 (5.8, 10.3) for those diagnosed between 1967 and 1976. New Zealanders with MS have a median survival age 7.2 years lower than the general population and twice the mortality risk. The survival gap was greater for progressive-onset disease and for those with an early age of onset.

Survival and cause of death in multiple sclerosis: results from a 50-year follow-up in Western Norway.

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Survival time among patients with multiple sclerosis (MS) has varied considerably according to previous reports. Survival and cause of death were analyzed among all patients with MS (878) with onset of MS in Hordaland County, Western Norway during 1953-2003, of whom 198 were dead at follow-up on January 1, 2005. Standardized mortality ratios (SMRs) and relative mortality ratios (RMRs) were calculated based on observed mortality in MS and expected mortality. Median survival from onset was 41 years versus 49 years in the corresponding population, and mortality (SMR) was 2.7-fold increased in MS. The median survival was 43 years among women and 36 years among men, but women had higher relative mortality, when compared with the corresponding population, than men (RMR = 1.40). The median survival time was 45 years among young-onset patients (21-30 years) and 23 years among older-onset patients (51-60 years), but young-onset patients had higher relative mortality than older-onset patients, as shown by a significant reduction by 10-year interval of age at onset (RMR = 0.65). Median survival from onset was longer (43 years) among relapsing-remitting MS than primary progressive MS ([PPMS]; 49 years), and the relative mortality was higher in the PPMS group, (RMR = 1.55). According to death certificates, 57% died from MS. Female patients and patients with young onset had longer median time to death but higher relative risk of dying compared with the corresponding population. PPMS had both shorter median time to death from onset and a higher relative risk of dying.

Survival of patients with multiple sclerosis in Denmark: a nationwide, long-term epidemiologic survey.

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We estimated survival probability and excess death rates for patients with MS on the basis of data from the Danish Multiple Sclerosis Registry, which includes virtually all patients diagnosed with MS in Denmark (population, five million) since 1948. We reviewed and reclassified all case records according to standardized diagnostic criteria. By linkage to the Danish Central Population Registry, we lost to follow-up only 25 patients who had emigrated. The median survival time from onset of the disease was 28 years in men (compared with 40 years in the matched general male population) and 33 years in women (versus 46 years). The median survival time from diagnosis was 22 years in men (versus 37 years) and 28 years in women (versus 42 years). The excess death rate between onset and follow-up (observed deaths per 1,000 person-years minus the expected number of deaths in a matched general population) was 14.3 in men, which was significantly higher than in women (12.0). Excess mortality increased with age at onset of MS in people of each sex. The 10-year excess death rate has decreased significantly in recent decades. Excess mortality was highest in cases with cerebellar symptoms at onset.

Prognostic factors for survival in multiple sclerosis.

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In a hospital-based study of 119 patients with definite multiple sclerosis, demographic and clinical factors were analysed with respect to their validity in assessing the long-term prognosis. Over a mean follow-up of 21.7 years, the following factors negatively influenced the prognosis by the univariate analysis: male sex, age at onset over 25, pyramidal involvement or spasticity at onset, > or =3 functional systems affected at onset or after 5 years, incomplete first remission, length of the first remission 5 attacks in the first 10 years, secondary or primary-progressive disease, time to reach secondary progression over 5 years and time to reach EDSS 6 over 7 years. The multivariate model showed that in patients with relapsing-remitting disease, 5 years after onset, pyramidal involvement at onset and shorter time to reach EDSS 6 predicted poor outcome, while after 10 years, higher age at onset and incomplete first remission indicated poor prognosis. Ten years after onset, the predictors of poor outcome in the secondary-progressive group were shorter time to reach EDSS 6 or secondary progression and higher EDSS, while in the primary-progressive group those variables were spasticity or higher number of functional systems affected at onset, and higher EDSS after 5 and 10 years.

Long-term prognosis of multiple sclerosis in Australia.

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Long-term survival and progression of disability in multiple sclerosis have not been studied previously in Australia. We report the findings in a cohort of 159 patients from Newcastle. Median survival time from onset of symptoms to death was 42 years. When expected survival rates are compared with those of the Australian population, there is approximately a 10% reduction in survival time in multiple sclerosis patients, after 20 years or more from disease onset. The expected time to reach DSS 3 and DSS 6 was 7 years and 27 years respectively. Survival time of multiple sclerosis patients and rate of progression of the disease are similar in the Northern and Southern Hemispheres.

Survival pattern and cause of death in patients with multiple sclerosis: results from an epidemiological survey in north east Scotland.

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The mean survival period in a series of 216 multiple sclerosis deaths, which formed part of a large prevalence sample observed in the Grampian region of Scotland, was 24.5 years, with an insignificant difference between females (25.7 years) and males (23.5 years). A third of the patients survived for over 30 years after onset. The age at death ranged between 25-80 years, with majority of the deaths occurring in the seventh decade (37%). On comparing life expectancy with the Scottish general population using life tables, only a slight reduction in the short-term (less than 10 years from onset) survival was noted in all age groups, with the exception of those with onset over the age of 50 years. The long-term life expectancy was however markedly reduced in all age groups compared with the controls. The survival period could be accurately predicted from the degree of disability at a point in time, and could be correlated with a number of clinical features, the most important of which was the age at onset. Eighty five per cent of those with onset of multiple sclerosis over the age of 50 years died within 20 years. Patients with a cerebellar disturbance at onset survived the shortest, and those with a brainstem lesion or retrobulbar neuritis the longest; those with a pyramidal dysfunction had an intermediate prognosis. Other parameters which could be correlated with the survival were: the timing and frequency of occurrence of psychiatric and urinary symptoms, interval between onset and first relapse and the course of the disease. As expected, most patients (89%) were significantly disabled (unable to walk) prior to death, only a minority, however, had become so within 10 years of the onset (10%). Sixty two per cent of the patients died of complications of multiple sclerosis. No unusual excess of any disease was noted amongst other causes. As expected, the majority of patients (55%) had bronchopneumonia as the terminal event, 11% had septicaemia, 15% had myocardial infarction and 4% had documented pulmonary embolism. This is the largest series of its kind where prognosis, judged by survival period, has been assessed amongst all multiple sclerosis patients derived from a prevalence sample and observed till death.